# Gojo 2020 pediatric ependymoma — analysis summary

RUN 1 OF 1

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

Harmony θ-sweep on sample: sample iLISI 1.5→6.6 (θ=10), one connected component (0 islands); PF/ST/SP subgroups grade continuously across one cloud (subgroup iLISI stays ~1.6 — biology preserved).

### Gate 2 — module geography → GO (non-branching)

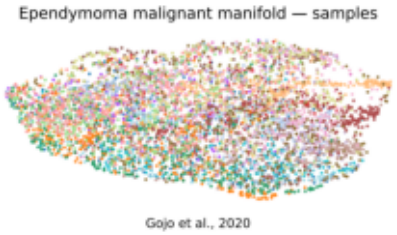
Non-branching: shared Gast/pSom body with an inverse terminal Crest gradient (Crest-high decile mean pseudotime = 0.97). Crest opposes Gast (r=−0.50) and pSom (r=−0.38); Gast~pSom weak (−0.20).

### Gate 3 — cross-repression → NA

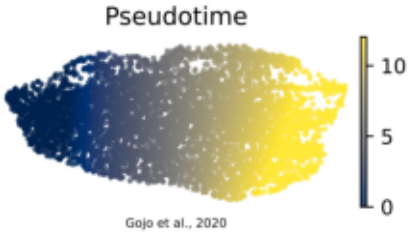
No competing Gast/pSom arms (weak −0.20, no bifurcation) — branching-only test does not apply.

## STEP FIGURES

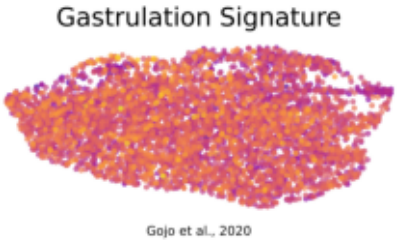
**Manifold — samples**



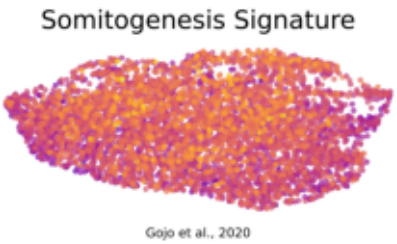

**Pseudotime**



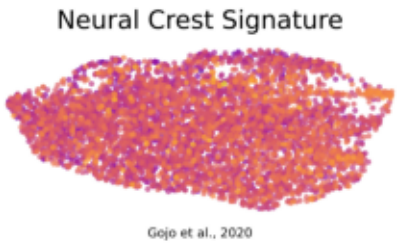

**Gastrulation**



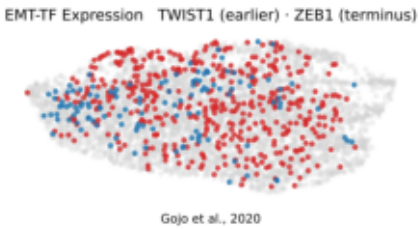

**Somitogenesis**




**Neural Crest**




**EMT-TF expression**




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**Dataset.**
Pediatric ependymoma (Gojo et al., 2020), SmartSeq2, 5,378 malignant cells across 20 samples (threshold ≥110/sample; PF/ST/SP molecular subgroups).

**Verdict.**
Pattern reproduces as the non-branching case — one continuous gastrulation/somitogenesis → neural-crest axis, no Gast/pSom bifurcation. Geometry: Linear-pSom/Crest. EMT-TFs overlay TWIST1 (earlier) ↔ ZEB1 (terminus).

## Parameter choices

| | | | |
|---|---|---|---|
| **Min-cell threshold** | ≥110 / sample (largest gap 140→88, geometric mean) | **Cells retained** | 5,551 → 5,378 (20/22 samples above threshold; 0 islands) |
| **Overdispersed genes** | 4,997 (scFates find_overdispersed) | **Integration** | Harmony on sample θ=10 (n_pcs=20, spread=1.3); iLISI 6.6 |
| **Geometry** | Non-branching; scFates curve N=30, λ=0.3, μ=100 (2 tips, 0 forks) | **Root** | Gast/pSom-high, Crest-low tip (Crest is late/terminal) |
| **Module means (raw)** | Gast +0.116 · pSom +0.329 · Crest +0.003 (pSom-dominant) | **EMT-TFs** | TWIST1 (earlier) ↔ ZEB1 (terminus); FOXC1/SNAI1/ZEB2/PRRX1 detected |